



UNIVERSITY OF PISA

Master of Science in Computer Engineering

DEPENDABLE AND SECURE SYSTEMS

ADVISE Car Attack

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Chapter 1

Introduction

The goal of this project is to create an ADVISE model on Mobius for adversaries trying to attack the communication channels of a car, using the threats suggested in Table [1.1](#).

1.1 Requirements

The requirements of this project are:

- It is required to build an attack graph with roughly 20 elements (knowledge, access, skill, attack step and goal).
- At least 2 different adversary profiles are required, varying initial elements and weights.
- The elements of the AEG must be explicitly linked with the vulnerabilities listed in Table [1.1](#).

1.2 Goals

The goals of this project are:

- Analyze and interpret the results.
- Evaluate the probability of each adversary to achieve the goal of stealing private data.

1.3 List of Vulnerabilities

E/ECE/TRANS/505/Rev.3/Add.154
Annex 5

Table A1
List of vulnerability or attack method related to the threats

High level and sub-level descriptions of vulnerability/ threat			Example of vulnerability or attack method	
4.3.2 Threats to vehicles regarding their communication channels	4	Spoofing of messages or data received by the vehicle	4.1	Spoofing of messages by impersonation (e.g. 802.11p V2X during platooning, GNSS messages, etc.)
			4.2	Sybil attack (in order to spoof other vehicles as if there are many vehicles on the road)
	5	Communication channels used to conduct unauthorized manipulation, deletion or other amendments to vehicle held code/data	5.1	Communications channels permit code injection , for example tampered software binary might be injected into the communication stream
			5.2	Communications channels permit manipulate of vehicle held data/code
			5.3	Communications channels permit overwrite of vehicle held data/code
			5.4	Communications channels permit erasure of vehicle held data/code
			5.5	Communications channels permit introduction of data/code to the vehicle (write data code)
	11	Messages received by the vehicle (for example X2V or diagnostic messages), or transmitted within it, contain malicious content	11.1	Malicious internal (e.g. CAN) messages
			11.2	Malicious V2X messages, e.g. infrastructure to vehicle or vehicle-vehicle messages (e.g. CAM, DENM)
			11.3	Malicious diagnostic messages
			11.4	Malicious proprietary messages (e.g. those normally sent from OEM or component/system/function supplier)

Figure 1.1: Vulnerabilities' Table

Chapter 2

Attack Execution Graph

This chapter presents the Attack Execution Graph (AEG) developed to model and analyze the cyber threats targeting the communication channels of a connected vehicle. The model has been implemented using the ADVISE formalism within the Mobius modeling environment.

2.1 AEG

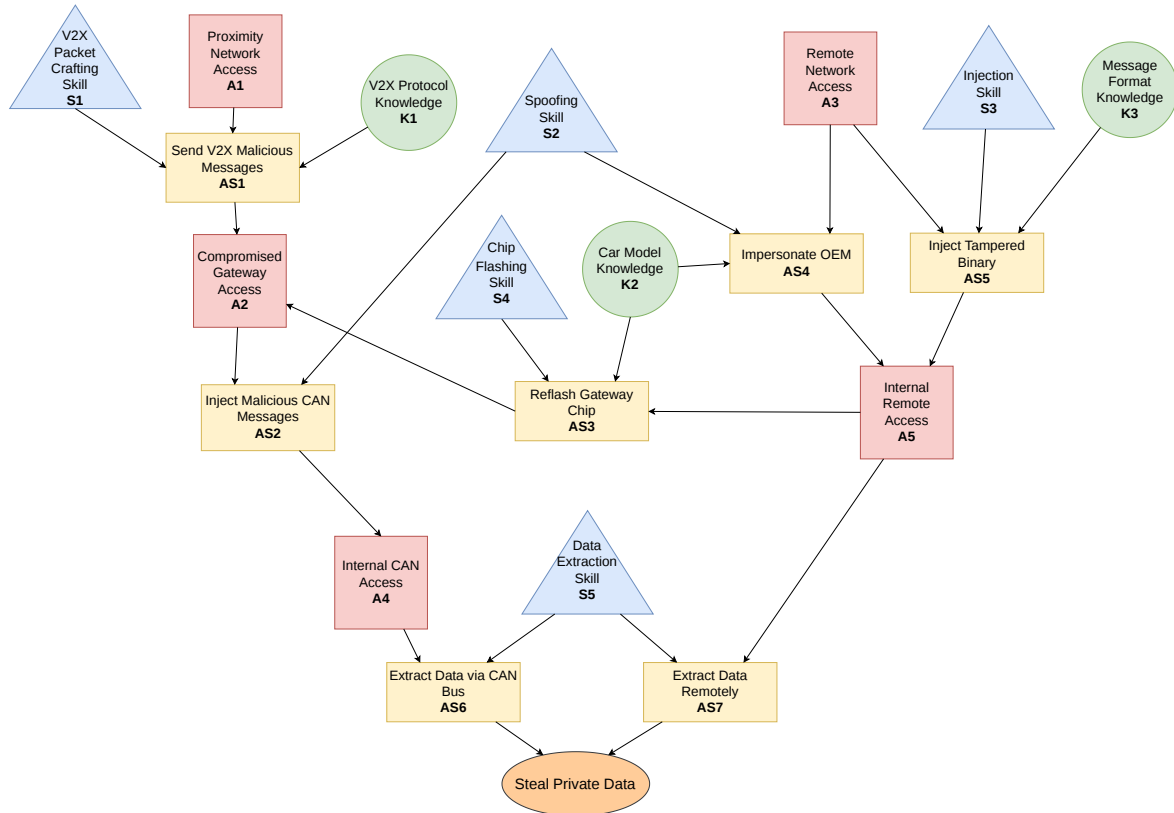


Figure 2.1: The Attack Execution Graph.

2.2 Elements

Here are the elements implemented in the model and their description.

2.2.1 Knowledge

- **K1: V2X Protocol Knowledge** → Knowledge regarding the V2X communication protocols used by the vehicle.
- **K2: Car Model Knowledge** → Knowledge about the specific model of the car being attacked.
- **K3: Message Format Knowledge** → Understanding of the internal message formats and structures (for example, diagnostic or proprietary messages).

2.2.2 Access

- **A1: Proximity Network Access** → Initial physical proximity allowing interception or transmission of short-range signals.
- **A3: Remote Network Access** → Access to the external communication channels of the vehicle from a remote location.

After-steps access

- **A2: Compromised Gateway Access** → Authenticated or compromised access to the gateway mediating the V2X network.
- **A4: Internal CAN Access** → Access to the internal Controller Area Network of the vehicle.
- **A5: Internal Remote Access** → Privilege to interact with internal remote services or endpoints of the car.

2.2.3 Skill

- **S1: V2X Packet Crafting Skill** → Ability to craft and manipulate V2X communication packets.
- **S2: Spoofing Skill** → Skill to forge messages and impersonate legitimate entities.
- **S3: Injection Skill** → Capability to inject tampered software binaries or code into communication streams.
- **S4: Chip Flashing Skill** → Ability to flash or rewrite the firmware of hardware chips and microcontrollers.
- **S5: Data Extraction Skill** → Advanced skill to parse, manipulate, and extract sensitive data from vehicle systems.

2.2.4 Attack Steps

Available steps, their effects, and their link to the threat table (1.1):

- **AS1: Send V2X Malicious Messages** → The adversary uses proximity access and V2X knowledge to craft and send malicious V2X messages.
 - *Table Link:* Section 11.2 (Malicious V2X messages).
- **AS2: Inject Malicious CAN Messages** → The adversary leverages a compromised gateway to spoof and inject malicious traffic directly into the CAN bus.
 - *Table Link:* Section 11.1 (Malicious internal (e.g. CAN) messages).
- **AS3: Reflash Gateway Chip** → Exploiting internal remote access, car model knowledge, and chip flashing skills to rewrite the gateway firmware, effectively compromising it.
 - *Table Link:* Section 5.3 (Communications channels permit overwrite of vehicle held data/code).
- **AS4: Impersonate OEM** → Remote access and car model knowledge allow the adversary to send malicious proprietary messages.
 - *Table Link:* Section 11.4 (Malicious proprietary messages sent from OEM).
- **AS5: Inject Tampered Binary** → The adversary injects malicious binaries through remote networks.
 - *Table Link:* Section 5.1 (Communications channels permit code injection, tampered software binary).
- **AS6: Extract Data via CAN Bus** → Data extraction through the compromised internal CAN bus.
 - *Table Link:* Section 5.2 (Communications channels permit manipulate of vehicle held data/code).
- **AS7: Extract Data Remotely** → Exploiting internal remote access to manipulate and exfiltrate data.
 - *Table Link:* Section 5.2 (Communications channels permit manipulate of vehicle held data/code).

2.2.5 Goal

- **G1: Steal Private Data** → Adversary's ultimate objective to exfiltrate private or sensitive information from the vehicle.

Chapter 3

Experiment Setup

3.1 Evaluation Scales

To perform the experiments, the following scales were defined for cost, time, payoff and skill of the hypothesized adversary.

3.1.1 Cost and Time Scale

This subsection defines the metrics used to quantify the financial resources (Table 3.1) and the estimated duration required to successfully execute an attack step (Table 3.2). One cost unit is equivalent to \$100, and one time unit corresponds to half an hour.

Range	Interpretation
0–20	Very cheap, affordable
21–50	Requires a small investment
51–80	Requires a serious budget
81–100	Very expensive

Table 3.1: Common reference scale for cost

Range	Interpretation
0–2	Requires one hour
3–5	Requires around two hours
6–8	Requires three to four hours
9–10	Requires up to five hours

Table 3.2: Common reference scale for time

3.1.2 Skill Scale

This subsection details the skill levels, categorizing the technical capabilities required by an attacker to exploit the vulnerabilities.

Range	Level	Interpretation
0–200	Novice	No specific skills
201–500	Intermediate	Security researcher
501–800	Advanced	Professional penetration tester
801–1000	Expert	World expert

Table 3.3: Common reference scale for adversary skills

3.1.3 Payoff Choice

The payoff of G1 was set to 1000, the maximum value allowed by Mobius: since it is the only goal in the model, the payoff plays no role in ordering alternative objectives, and the maximum value is consistent with the cost scale (0–100) defined within the tool itself.

3.2 Attack Step Configurations

This section specifies the quantitative parameters assigned to each attack step within the ADVISE model, including required skills, execution costs, and probabilities.

3.2.1 Skill Thresholds

The following table and descriptions define the minimum skill level required to attempt each specific attack step.

Attack Step	Required Skill	Threshold
AS1 – Send V2X Malicious Messages	S1 – V2XPacketCraftingSkill	300
AS2 – Inject Malicious CAN Messages	S2 – SpoofingSkill	350
AS3 – Reflash Gateway Chip	S4 – ChipFlashingSkill	700
AS4 – Impersonate OEM	S2 – SpoofingSkill	450
AS5 – Inject Tampered Binary	S3 – InjectionSkill	500
AS6 – Extract Data via CAN Bus	S5 – DataExtractionSkill	250
AS7 – Extract Data Remotely	S5 – DataExtractionSkill	300

Table 3.4: Required skill thresholds for each attack step

3.2.2 Execution Parameters and Probabilities

Attack Step	Cost	Time	Outcome Prob.		Detection Prob.	
			Success	Failure	Success	Failure
AS1 – Send V2X Malicious Messages	10	3	0.40	0.60	0.35	0.45
AS2 – Inject Malicious CAN Messages	10	2	0.75	0.25	0.15	0.20
AS3 – Reflash Gateway Chip	35	9	0.70	0.30	0.30	0.30
AS3 – Reflash Gateway Chip – EXP	15	5	0.90	0.10	0.10	0.25
AS4 – Impersonate OEM	30	5	0.70	0.30	0.10	0.10
AS5 – Inject Tampered Binary	20	7	0.75	0.25	0.30	0.30
AS6 – Extract Data via CAN Bus	5	2	0.90	0.10	0.10	0.30
AS7 – Extract Data Remotely	15	3	0.50	0.50	0.30	0.60

Table 3.5: Cost, Time, and Probabilities of Attack Steps.

This subsection presents the complete set of parameters for each attack step, detailing the associated costs, execution times, and the probabilities of success and detection.

The attack step AS3 – **EXP** refers to the scenario where the relevant skill exceeds the specific threshold of 800, classifying the adversary as an expert. Consequently, both the cost and time required for the attack decrease, while the probability of success increases significantly.

3.3 Adversary Modeling

This section introduces the variables and profiles used to simulate different types of attackers and test the execution graph against varying initial capabilities.

3.3.1 Global Variables

In order to simulate different adversary profiles and control the initial Knowledge, Access, and Skill elements, 10 global variables were defined in the ADVISE model.

Type	Global Variable	Value
bool	A1 – hasProximityAccess	[0, 1]
bool	A3 – hasRemoteNetworkAccess	[0, 1]
bool	K1 – hasV2XProtocolKnowledge	[0, 1]
bool	K2 – hasCarModelKnowledge	[0, 1]
bool	K3 – hasMessageFormatKnowledge	[0, 1]
short	S1 – v2XPacketCraftingSkillLevel	[0, 1000]
short	S2 – spoofingSkillLevel	[0, 400, 1000]
short	S3 – injectionSkillLevel	[0, 1000]
short	S4 – chipFlashingSkillLevel	[0, 750, 1000]
short	S5 – dataExtractionSkillLevel	[0, 275, 1000]

Table 3.6: Adversary Capability Attributes in the ADVISE Model

The total number of possible configurations amounts to 3456. However, only the most significant experiments, presented in Section 3.3.3, were taken into consideration.

3.3.2 Adversary Profiles

The possible global variable configurations can be grouped into three major categories based on the adversary’s initial access to the system. Configurations that prevent the adversary from executing any attack, due to insufficient access or skills, were excluded from the analysis.

- **Remote Adversary:** This profile models an attacker starting exclusively with *Remote Network Access* (A3).
- **Proximity Adversary:** This profile represents an actor operating within the physical vicinity of the vehicle, starting with *Proximity Network Access* (A1).
- **Hybrid Adversary:** This profile models a highly equipped actor possessing both *Proximity* (A1) and *Remote Network Access* (A3).

Adversary Weights

To simulate various decision-making processes during the evaluation of attack steps, each experiment was performed applying different weights to cost, detection, and payoff, reflecting distinct attacker behaviors:

Mindset	Cost Weight	Detection Weight	Payoff Weight
Economic	0.4	0.1	0.5
Stealthy	0.1	0.3	0.6
Stubborn	0.1	0.1	0.8

Table 3.7: Adversary weights

3.3.3 Experiments

As previously stated, only a select number of significant configurations was evaluated. Each scenario was executed in a 150-time-unit window, giving plenty of time for the attacker to reach the goal, as the primary focus was to analyze path preferences and execution efficiency rather than evaluating the adversary under strict time constraints. A total of 7 experiments were performed, categorized into 4 subgroups. For each subgroup containing a pair of experiments, the first configuration models a baseline attacker, while the second simulates an attacker with a chip flashing skill above the expert threshold.

- **Remote Adversary (Experiments 3419, 3420):** These experiments model a remote adversary, whose available attack paths are shown in Figure 3.1.
- **CAN-Bound Adversary (Experiments 3452, 3453):** These experiments model a hybrid adversary who lacks the required skill to exfiltrate data remotely. The available attack paths are shown in Figure 3.2.
- **No Flashing Adversary (Experiment 3454):** This experiment models a hybrid adversary who lacks the required skill to flash the gateway chip. The available attack paths are shown in Figure 3.3.
- **Complete Adversary (Experiments 3455, 3456):** These experiments model an adversary with complete access to all attack paths, as illustrated in Figure 3.4.

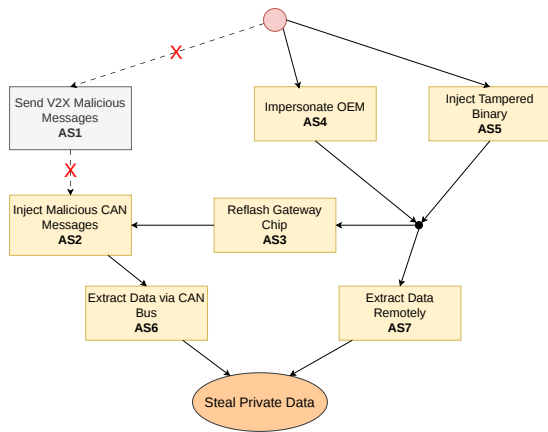


Figure 3.1: Remote Adversary Only.

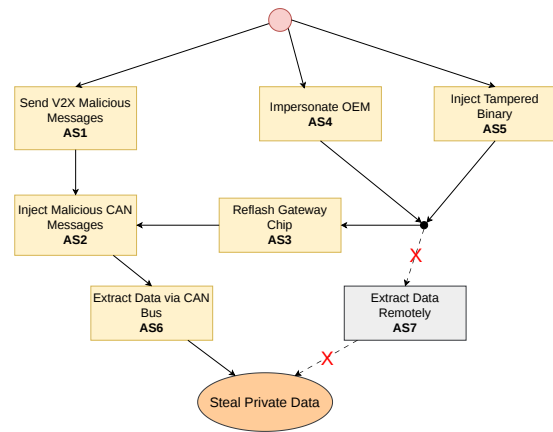


Figure 3.2: Lack of extraction skill.

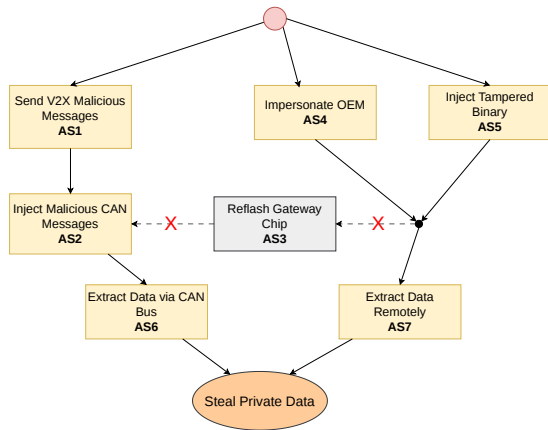


Figure 3.3: Lack of chip flashing skill.

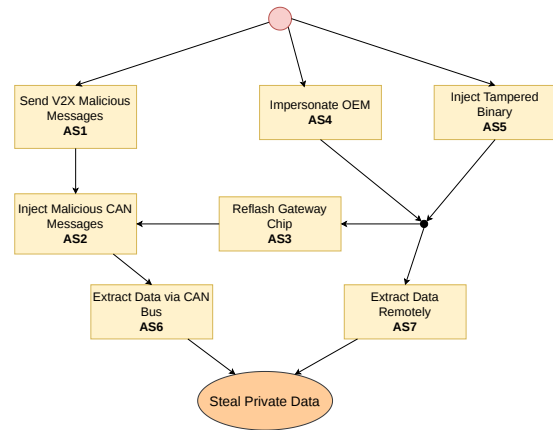


Figure 3.4: All paths available.

Chapter 4

Results

4.1 Payoff-Based Adversary

This section presents the results for adversaries operating with a payoff-oriented mindset, prioritizing the successful exfiltration of data, while placing less weight on attack costs or stealth.

Cost Weight	Detection Weight	Payoff Weight
0.1	0.1	0.8

4.1.1 Simulation Results

Table 4.1 summarizes the simulation results, detailing the chosen attack path, the initial time unit at which the probability of data exfiltration becomes strictly positive ($T_{P>0}$), and the exact moment it reaches absolute certainty ($T_{P=1}$).

Experiment	Chosen Attack Path	$T_{P>0}$	$T_{P=1}$
Exp 3419	Inject Binary $\rightarrow$ Extract Remotely	10	62
Exp 3420	Inject Binary $\rightarrow$ Reflash $\rightarrow$ Inject CAN $\rightarrow$ Extract CAN	16	53
Exp 3452	Send V2X $\rightarrow$ Inject CAN $\rightarrow$ Extract CAN	7	53
Exp 3453	Send V2X $\rightarrow$ Inject CAN $\rightarrow$ Extract CAN	7	53
Exp 3454	Inject Binary $\rightarrow$ Extract Remotely	10	62
Exp 3455	Inject Binary $\rightarrow$ Extract Remotely	10	62
Exp 3456	Inject Binary $\rightarrow$ Reflash $\rightarrow$ Inject CAN $\rightarrow$ Extract CAN	16	53

Table 4.1: Simulation results for each experiment (Payoff Mindset).

4.1.2 Path Analysis

- **Remote Adversary (Exp 3419, 3420):** The baseline remote attacker (Exp 3419) naturally favors the most direct remote approach (*Inject Tampered Binary $\rightarrow$ Extract*

Data Remotely), as referenced in Figure 4.1. However, when the attacker possesses expert-level chip flashing skills (Exp 3420), the preferred path shifts to compromising the gateway (*Inject Tampered Binary* → *Reflash Gateway Chip* → *Inject Malicious CAN Messages* → *Extract Data via CAN Bus*), illustrated in Figure 4.2. The increased success probability of the expert flashing step make entering the internal CAN bus a more lucrative and reliable choice than purely remote extraction.

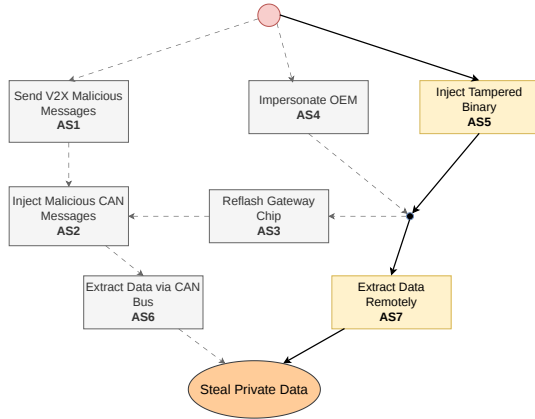


Figure 4.1

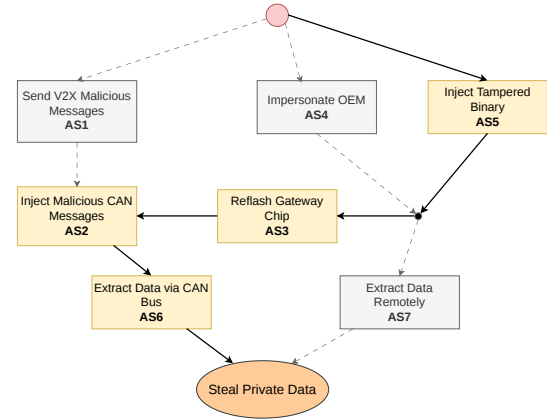


Figure 4.2

- **CAN-Bound Adversary (Exp 3452, 3453):** Lacking remote extraction capabilities, these adversaries must rely on proximity access. Both the baseline and expert configurations opt for the exact same path (*Send V2X Malicious Messages* → *Inject Malicious CAN Messages* → *Extract Data via CAN Bus*), shown in Figure 4.3.

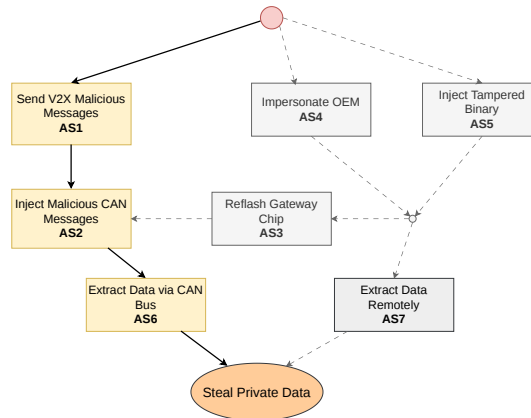


Figure 4.3

- **No Flashing Adversary (Exp 3454):** Without the ability to reflash the gateway chip, the hybrid attacker defaults to the purely remote path (*Inject Tampered Binary* → *Extract Data Remotely*) (Figure 4.1).
- **Complete Adversary (Exp 3455, 3456):** When given unrestricted access, the baseline attacker (Exp 3455) still prefers the remote injection and extraction path (Figure 4.1), confirming it as the path of least resistance for a standard payoff-driven

attacker. Conversely, the expert adversary (Exp 3456) prefers to use their advanced skill to compromise the gateway and operate directly within the CAN bus (Figure 4.2).

4.2 Cost-Based Adversary

This section analyzes the results for adversaries operating with an economic mindset. These attackers aim to achieve their goal with the minimum possible financial investment, placing a higher weight on cost reduction rather than stealth or execution speed.

Cost Weight	Detection Weight	Payoff Weight
0.4	0.1	0.5

4.2.1 Simulation Results

Table 4.2 summarizes the simulation results, detailing the chosen attack path, the initial time unit at which the probability of data exfiltration becomes strictly positive ($T_{P>0}$), and the exact moment it reaches absolute certainty ($T_{P=1}$).

Experiment	Chosen Attack Path	$T_{P>0}$	$T_{P=1}$
Exp 3419	Inject Binary $\rightarrow$ Extract Remotely	10	62
Exp 3420	Inject Binary $\rightarrow$ Extract Remotely	10	62
Exp 3452	Send V2X $\rightarrow$ Inject CAN $\rightarrow$ Extract CAN	7	53
Exp 3453	Send V2X $\rightarrow$ Inject CAN $\rightarrow$ Extract CAN	7	53
Exp 3454	Send V2X $\rightarrow$ Inject CAN $\rightarrow$ Extract CAN	7	53
Exp 3455	Send V2X $\rightarrow$ Inject CAN $\rightarrow$ Extract CAN	7	53
Exp 3456	Send V2X $\rightarrow$ Inject CAN $\rightarrow$ Extract CAN	7	53

Table 4.2: Simulation results for each experiment (Economic Mindset).

4.2.2 Path Analysis

When observing the execution paths selected under the economic mindset, a clear pattern emerges based on the total cost of the attack steps:

- **Remote Adversary (Exp 3419, 3420):** Lacking physical proximity, these adversaries are restricted to remote entry points. Both the baseline (Exp 3419) and the expert attacker (Exp 3420) select the *Inject Tampered Binary $\rightarrow$ Extract Data Remotely* path (Figure 4.4), which carries a cumulative cost of 35. Even though the expert attacker possesses advanced chip flashing skills that lower the cost of the reflash step from 35 to 15, the overall cost of the gateway compromise path would still be higher than the chosen one.

- **Proximity-Enabled Adversaries (Exp 3452, 3453, 3454, 3455, 3456):** Every configuration possessing *Proximity Network Access*, regardless of their other skill levels or limitations, converges on the exact same path: *Send V2X Malicious Messages* → *Inject Malicious CAN Messages* → *Extract Data via CAN Bus* (Figure 4.5). This makes the selected path the absolute cheapest attack across the entire Attack Execution Graph.

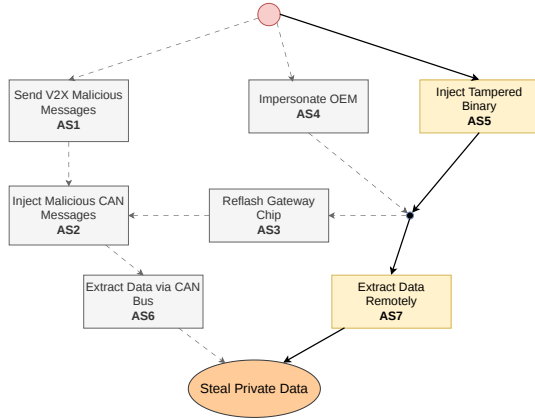


Figure 4.4

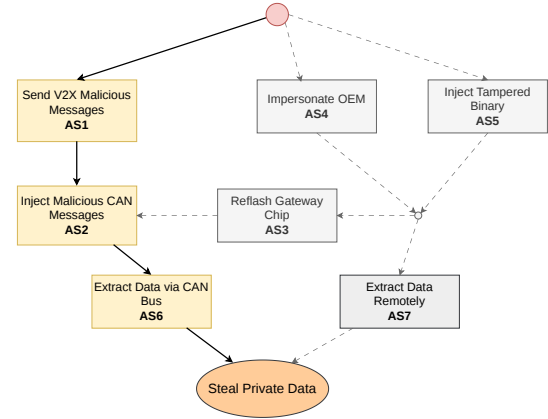


Figure 4.5

4.3 Detection-Based Adversary

This section details the results for adversaries operating with a stealthy mindset. These attackers prioritize avoiding detection at all costs, selecting paths with the lowest probability of triggering alarms, even if they require more time or financial resources.

Cost Weight	Detection Weight	Payoff Weight
0.1	0.3	0.6

4.3.1 Simulation Results

Table 4.3 summarizes the simulation results, detailing the chosen attack path, the initial time unit at which the probability of data exfiltration becomes strictly positive ($T_{P>0}$), and the exact moment it reaches absolute certainty ($T_{P=1}$).

Experiment	Chosen Attack Path	$T_{P>0}$	$T_{P=1}$
Exp 3419	Impersonate OEM $\rightarrow$ Extract Remotely	8	38
Exp 3420	Impersonate OEM $\rightarrow$ Reflash $\rightarrow$ Inject CAN $\rightarrow$ Extract CAN	14	54
Exp 3452	Send V2X $\rightarrow$ Inject CAN $\rightarrow$ Extract CAN	7	53
Exp 3453	Impersonate OEM $\rightarrow$ Reflash $\rightarrow$ Inject CAN $\rightarrow$ Extract CAN	14	54
Exp 3454	Impersonate OEM $\rightarrow$ Extract Remotely	8	38
Exp 3455	Impersonate OEM $\rightarrow$ Extract Remotely	8	38
Exp 3456	Impersonate OEM $\rightarrow$ Reflash $\rightarrow$ Inject CAN $\rightarrow$ Extract CAN	14	54

Table 4.3: Simulation results for each experiment (Stealthy Mindset).

4.3.2 Path Analysis

The introduction of a high detection weight drastically alters the execution paths compared to the payoff and cost-oriented mindsets. The model systematically avoids steps with high detection probabilities:

- **Remote Adversary (Exp 3419, 3420):** The baseline attacker (Exp 3419) abandons the noisy binary injection in favor of *Impersonate OEM $\rightarrow$ Extract Data Remotely* (Figure 4.6). Impersonating the OEM carries a significantly lower detection probability, making it the optimal choice for a stealthy baseline approach. Conversely, the expert attacker (Exp 3420) opts for a hardware-level compromise (*Impersonate OEM $\rightarrow$ Reflash Gateway Chip $\rightarrow$ Inject Malicious CAN Messages $\rightarrow$ Extract Data via CAN Bus*), shown in Figure 4.7. The expert threshold drops the detection probability of the flashing step to a mere 0.10, making this complex path stealthier than purely remote extraction.

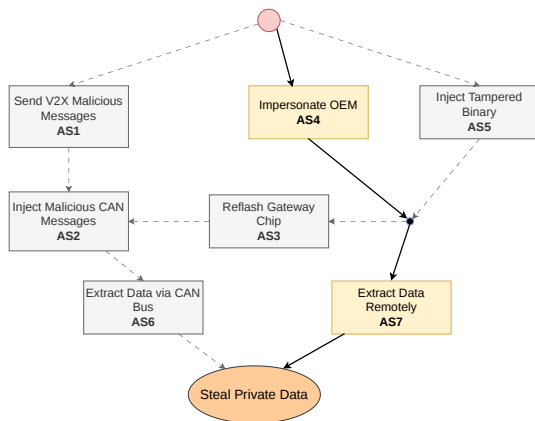


Figure 4.6

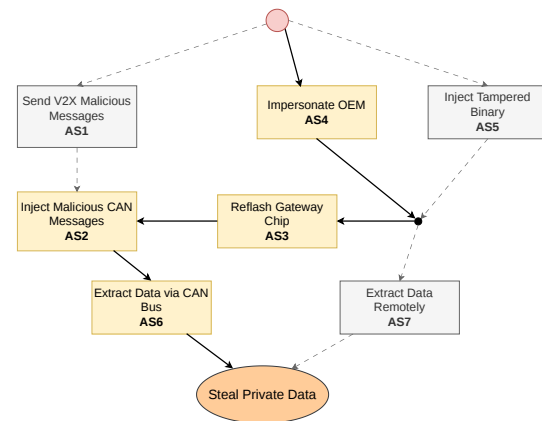


Figure 4.7

- **CAN-Bound Adversary (Exp 3452, 3453):** The baseline attacker (Exp 3452) lacks both remote extraction and advanced chip-flashing capabilities, leaving the highly detectable V2X entry, shown in Figure 4.8, as the only physically possible path

to reach the goal. However, the expert attacker (Exp 3453) utilizes their advanced flashing skills to completely bypass the noisy V2X vulnerability, executing a discreet hardware compromise shown in Figure 4.7 (*Impersonate OEM* $\rightarrow$ *Reflash Gateway Chip* $\rightarrow$ *Inject Malicious CAN Messages* $\rightarrow$ *Extract Data via CAN Bus*).

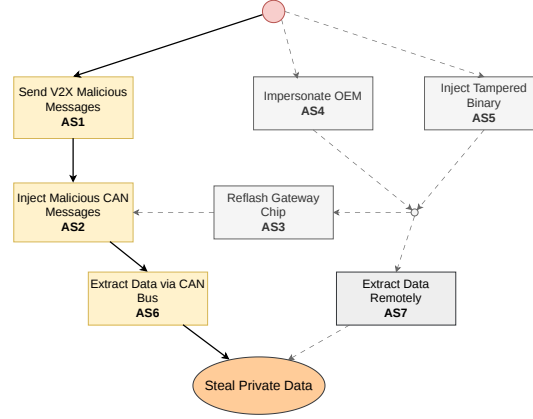


Figure 4.8

- **No Flashing Adversary (Exp 3454):** Without the ability to reflash the gateway chip, the hybrid attacker defaults to the safest purely remote path: *Impersonate OEM* $\rightarrow$ *Extract Data Remotely* (Figure 4.6), avoiding the highly detectable proximity-based steps.
- **Complete Adversary (Exp 3455, 3456):** When given unrestricted access, the baseline attacker (Exp 3455) still prefers *Impersonate OEM* $\rightarrow$ *Extract Data Remotely* (Figure 4.6) as the most effective route. Meanwhile, the expert adversary (Exp 3456) leverages their advanced skills to execute the hardware-level compromise (*Impersonate OEM* $\rightarrow$ *Reflash Gateway Chip* $\rightarrow$ *Inject Malicious CAN Messages* $\rightarrow$ *Extract Data via CAN Bus*) (Figure 4.7), confirming it as the absolute optimal path for a stealth-driven expert attacker.

Chapter 5

Conclusions

This project successfully developed and evaluated an ADVISE model within the Mobius environment to analyze cyber threats targeting the communication channels of a connected vehicle. The framework assesses how different adversary profiles operate, depending on their initial access, technical expertise, and strategic priorities.

5.1 Key Findings

The execution of the simulations revealed clear behavioral patterns depending on the attacker's priorities:

- **Payoff Optimization:** When the objective is maximizing the reward and success rate, adversaries naturally prefer the path of least resistance, such as direct remote extraction. However, the results demonstrated that reaching an expert level in complex hardware manipulation (gateway reflashing) fundamentally alters the optimal strategy, shifting the attacker's focus toward a deep compromise of the internal CAN bus, which becomes the most reliable method for data exfiltration.
- **Cost Minimization:** The economic mindset highlighted the main problem of proximity based interfaces. Regardless of other available advanced capabilities, cost-driven attackers consistently chose to exploit V2X communication channels. The low financial barrier associated with sending malicious V2X messages makes the physical vicinity of the car the most accessible and cost-effective entry point.
- **Stealth:** Prioritizing detection avoidance drastically shifted the selected attacks. The model captured the adversaries' reluctance to use "noisy" methods like V2X manipulation or binary injections, instead relying on the impersonation of the OEM. Furthermore, expert hardware skills proved invaluable once more, allowing attackers to ensure a minimal probability of detection.

5.2 Final Remarks

Ultimately, these findings demonstrate that real-world exploitation is fundamentally driven by the adversary's constraints (budget, time) and strategic goals (reward, stealth). To

effectively protect connected vehicles, manufacturers must adopt a strict defensive approach. This strategy requires not only securing accessible external interfaces, such as V2X and remote endpoints, but also implementing robust internal monitoring to help neutralize highly skilled attackers who manage to bypass the initial perimeter. Using better components and protocols can also be a valid strategy, as it may increase the cost and difficulty of most attacks, while also reducing the adversary's success probabilities.